Final, CS-GY 6923

Sample Questions

Show all of your work to receive full (and partial) credit.

Always, Sometimes, Never

Indicate whether each of the following statements is **always** true, **sometimes** true, or **never** true. Provide a one or two short justification or example to explain your choice.

1. Given a linearly separable data set, an optimal solution to the soft-margin SVM objective will be a correct separating hyperplane for the dataset.

   ALWAYS **SOMETIMES** NEVER

   The soft-margin SVM allows some points to be misclassified, even if the data set is linearly seperable. If we set C large enough, however, a soft-margin SVM converges to a hard margin classifier, so we will get a seperating hyperplane. See slide 59 here: https://www.chrismusco.com/machinelearning2023_grad/lectures/lec8.pdf.

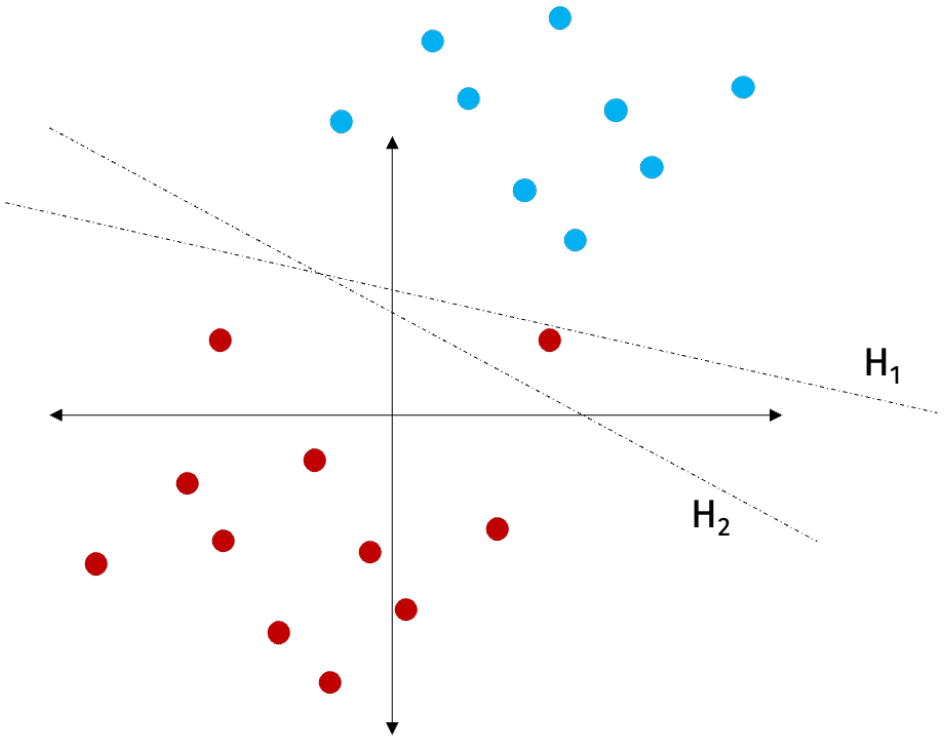
2. Let K be a kernel gram matrix generated from a datasets $x_1, \ldots, x_n$ and a PSD kernel function k. K has a symmetric Signular Value Decomposition (meaning the same left and right singular vectors).

   ALWAYS SOMETIMES NEVER

   If K is PSD, then by definition it can be written as $\Phi\Phi^T$ for some matrix Φ containing feature transformations of $x_1, \ldots, x_n$. Let $U\Sigma V^T = \Phi$ be the singular value decompostion of Φ. Then we have that $K = U\Sigma V^T (U\Sigma V^T)^T = U\Sigma V^T V \Sigma U^T = U\Sigma^2 U^T$. In the last step I used that V is orthogonal so $V^T V = I$. So, we see by observation that K can be written as the product of U, a positive diagonal matrix Σ^2, and U^T, so it has a symmetric SVD.

Short Answer

3. In the plot below, H_1 and H_2 are hyperplanes obtained by trainging a soft-margin SVM with different values of C. Which one was trained with a larger value of C? On the same plot draw the hyperplane that you believe would be returned by a hard margin SVM.



H_1 was likely trained with a larger value of C, since its solution has few total misclassified points. H_1 more closely approximates the hard margin classifier. H_2 was likely trained with the smaller value of C

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4. **TRUE** or FALSE. A convolutional layer that takes in $n \times n$ images and processes them with a series of $n \times n$ convolutional filters is equivalent to a fully connected layer.

   Convolving and $n \times n$ matrix with an $n \times n$ filter leads to a 1 dimensional output, which is just an arbitrary sum of the values in the input. So this is equivalent to a fully connected layer where the number of hidden neurons equals the number of $n \times n$ convolutional filters.

5. An alternative definition of a PSD kernel function that you will see in many text books is as follows: We say that k is PSD if for *any* dataset $x_1, \ldots, x_n$, the kernel gram matrix K with $K_{ij} = k(x_i, x_j)$ is ``positive semi-definite'', where we say a matrix K is positive semidefinite if, for *all* vectors x, $x^T K x \geq 0$ (this is a definition you might have seen before in a linear algebra class).

   Prove that the other definition we gave for k (i.e. that $k(x_i, x_j) = \phi(x_i)^T \phi(x_j)$ for some feature transformtion ϕ Implies this definition (you don't need to show the other way).

   Given the definition from class, we know that $K = \Phi\Phi^T$ for some matrix Φ. so $x^T K x = x^T \Phi \Phi^T x = \|\Phi^T x\|_2^2$. The squared two norm of a vector is always ≥ 0, so our definition implies that $x^T K x \geq 0$ as desired.

6. Draw a linearly seperable data set where there is no unique maximum margin classifier. I.e. a hard margin SVM could return multiple different solutions.

7. As discussed in class, convolutional networks can be viewed as special cases of fully connected networks with weight sharing. Of course, weight sharing effects how gradients of computed for the neural network. Revisit Problem 3(d) on Homework 3 and recommute the gradient for the network where we have the additional weight sharing constraints that $W_{H,2} = W_{H,3}$ and $W_{O,1} = W_{O,2}$.